Reading Poverty from AlphaEarth: Augmented Multidimensional Prediction and a Self-Supervised Poverty Representation

Preliminary results · work in progress · V1

Abstract

This paper builds a seven-country **H3 resolution-7** tabular spine for multidimensional poverty prediction by combining Google AlphaEarth embeddings with population, nightlights, vegetation, building, road, labelled with DHS-derived outcomes. The current analysis covers Ghana, Gambia, Kenya, Nigeria, Sierra Leone, Tanzania, and Zambia *(add more training cases in v2)*. After spatial-block buffering, the effective labeled working set is **4,266 H3 resolution-7 cells** (4,219 for stunting), drawn from 5,186 total labeled cells.

The following questions are of interest: **(i) how well do AlphaEarth embeddings predict five DHS socioeconomic outcomes, and does augmenting them with additional socioeconomic information derived purely from Earth-observation sources further improve performance**, evaluated under strict spatial-block cross-validation and leave-one-country-out transfer; **(ii) how do different model families compare**—gradient boosting versus a graph neural network over H3 adjacency; and **(iii) can a self-supervised fused embedding, trained with no DHS information, reach comparable accuracy on the same prediction tasks**.

Results indicate that **AlphaEarth carries useful socioeconomic signal**, strongest for wealth and women's education, and that **explicit EO-derived socioeconomic features produce consistent spatial-CV improvements** for wealth, sanitation, women's education, and mobile ownership; the improvement is small for stunting. For wealth, the augmented model reaches pooled $R^2 = 0.750$ under spatial cross-validation and 0.673 under leave-one-country-out transfer. These estimates fall within a similar range to established EO-poverty benchmarks, providing an encouraging direction for the augmented AlphaEarth approach, while remaining indicative rather than directly equivalent because of methodological differences including country coverage and spatial unit. A single-task H3 resolution-7 GraphSAGE model is broadly comparable with gradient boosting on the full feature matrix, without a consistent advantage across outcomes. The **self-supervised fused embedding—trained without any DHS labels**—retains useful wealth and sanitation signal, although it remains below the full supervised models. Leave-one-country-out transfer is strongest for wealth and sanitation, positive but weaker for women's education and mobile ownership, and near zero for stunting.

1 Motivation And Prior Work

The broader project asks how far Earth observation and AI can go in estimating multidimensional poverty over space and time, and when a cheaper representation is good enough relative to raw satellite imagery. This version focuses on a narrower step: **can AlphaEarth be made more poverty-aware by adding interpretable, scalable geospatial features at the H3 resolution-7 level?** The contribution being tested is not that population density, roads, nightlights, or building counts are new. The contribution is the augmentation and evaluation design: AlphaEarth alone, tabular features alone, AlphaEarth plus tabular groups, graph models, and a DHS-free fused embedding.

The older satellite-poverty papers give a scale for what "good" has meant in related work, while the newer representation papers explain why a compact-embedding route is worth testing. Jean et al. established the satellite-to-poverty template by using imagery and machine learning to infer survey wealth. Yeh et al. scaled that logic with public imagery across ∼20,000 African villages, reporting a held-out-country wealth R^2 of 0.67 on pooled cluster-level observations (or 0.70 when averaging per-country R^2). Pettersson et al. moved the problem toward temporal EO representation learning, using time series of satellite imagery across a larger DHS panel and reporting 72% of held-out-country and 75% of held-out-time variance explained for wealth. These papers motivate the benchmark question for this version: whether an H3 resolution-7 AlphaEarth-plus-tabular pipeline can approach the same wealth-prediction range while using precomputed embeddings and engineered socioeconomic layers rather than training a direct image sequence model.

The representation papers motivate a second question: whether the features can become a reusable embedding rather than just a supervised predictor. AlphaEarth Foundations motivates the 64-dimensional annual embedding as a compact general EO representation. Pettersson and Daoud use AlphaEarth with graph models for cluster-level poverty mapping across 37 DHS surveys and find that graph structure adds only a modest gain (best $R^2 = 0.569$ versus an image-only baseline of 0.546). Google PDFM and UrbanFusion are broader multimodal representation papers, not DHS poverty papers, but their methods are relevant because they combine spatial, human, environmental, and map-like modalities into reusable representations. MOSAIKS is an older precedent for shared satellite encodings with cheap downstream probes. In this paper, the frozen fused embedding is therefore read against the non-embedding models: XGBoost and GNN show what can be extracted with supervised access to the full feature matrix, while the embedding probe asks what survives when DHS is not used to train the representation.

2 Data Construction

2.1 H3 Resolution-7 Spine, Countries, And Labels

The unit of analysis is an **H3 resolution-7 cell**. The GEE spine generates one row per H3 resolution-7 cell with country, H3 index, centroid coordinates, area, reference year, and polygon geometry. Admin-1 labels are assigned by centroid. This keeps the table simple and model-ready, but border cells should be audited if admin-level aggregation becomes part of the final claims.

The current analytic table contains **657,863 H3 resolution-7 cells** across seven countries. DHS labels are sparse relative to the full grid: 5,186 H3 resolution-7 cells have wealth, sanitation, women's education, and mobile labels, and 5,132 have stunting labels. Ethiopia and Togo were present in upstream GEE feature generation but are not part of the current modeling set.

2.2 Source-Specific Feature Construction

The feature pipeline has two stages. The first builds the core satellite/composite features from Google Earth Engine: AlphaEarth, VIIRS, MODIS NDVI, and GHSL building height. The second processes WorldPop rasters, DHS labels, RWI audit fields, OSM roads, and Google Open Buildings morphology, then merges everything into a single H3 resolution-7 table.

AlphaEarth. AlphaEarth is treated as the core EO representation, not as a point sample. The extraction uses the annual `GOOGLE/SATELLITE_EMBEDDING/V1/ANNUAL` collection, top-coded to the latest available year when the DHS reference year is beyond source availability. The key implementation choice is polygon mean over each H3 resolution-7 cell rather than centroid pixel sampling. A centroid pixel would let a tiny point represent a multi-square-kilometer H3 resolution-7 cell; polygon means are slower but better aligned with the prediction unit. Chunked Drive exports and a 30 m reduction scale make this feasible.

- **Calculated features:** 64 H3 resolution-7 polygon-mean dimensions, `ae_A00`–`ae_A63`, plus AlphaEarth effective year.
- **Orthogonal features:** none. AlphaEarth is the base representation being augmented, not part of the orthogonal tabular block.

Nightlights. VIIRS is summarized as a 24-month signature ending in the reference year. Monthly images are masked by

Table 1: Current analytic country set and DHS-labeled H3 resolution-7 cells.

Country	H3 res-7	Wealth	Sanitation	Women edu.	Stunting	Mobile
Ghana	60,433	602	602	602	600	602
Gambia	2,316	172	172	172	172	172
Kenya	106,744	1,490	1,490	1,490	1,489	1,490
Nigeria	190,814	1,350	1,350	1,350	1,313	1,350
Sierra Leone	17,349	464	464	464	454	464
Tanzania	158,287	596	596	596	592	596
Zambia	121,920	512	512	512	512	512
Total	657,863	5,186	5,186	5,186	5,132	5,186

coverage, but raw radiance is not floored at zero before the mean and raw volatility are computed. This matters in dark rural areas, where small negative radiance values can carry calibration/background information. Log features use a tiny lower clamp only because logs require it.

- **Calculated features:** 24-month radiance mean, raw volatility, log slope, log volatility, valid-month count, and effective year.
- **Orthogonal features:** log slope and volatility. The slope is estimated over the 24-month log-radiance sequence; volatility captures instability rather than brightness level. Mean radiance is treated as an amount feature.

NDVI. NDVI was moved from Sentinel-2 to MODIS native 16-day products. Sentinel-2 had better spatial resolution, but for nine-country extraction it was slower, cloud-fragile, and memory-heavy in GEE. MODIS is coarser, but the denser annual sequence gives more stable phenology summaries.

- **Calculated features:** annual mean NDVI, robust seasonal amplitude, peak timing, valid-period counts, reliability flags, and effective year from 23 native 16-day periods.
- **Orthogonal features:** seasonal amplitude and peak timing. Amplitude is p90–p10 over the 16-day values rather than max–min, reducing sensitivity to single bad observations. Peak timing approximates when greenness is highest; NDVI mean is the amount-like vegetation feature.

WorldPop. WorldPop is processed locally from 100 m rasters. Local processing is used because raster nodata/sentinel values need explicit masking, and because rasterize-once plus bincount aggregation is much faster than polygon-by-polygon zonal statistics over many rasters and many H3 resolution-7 cells.

- **Calculated features:** population count, population density, youth dependency ratio, elderly dependency ratio, and sex ratio.
- **Orthogonal features:** youth dependency, elderly dependency, and sex ratio. These are derived from age-sex raster sums and are intended to capture demographic composition, not settlement amount. Population count/density are amount features.

GHSL building height. GHSL height is a static 2018 layer in the selected source. It is therefore treated as a structural proxy rather than a year-matched DHS feature.

- **Calculated features:** mean, p90, and max built height within each H3 resolution-7 cell.
- **Orthogonal features:** height distribution. Verticality is not the same as footprint count; p90 and max height help distinguish dense low-rise settlement from denser vertical settlement.

Google Open Buildings. Google Open Buildings is filtered at confidence ≥ 0.70. For Gambia, Ghana, Kenya, Sierra Leone, and Zambia, the direct Open Buildings S2 shard download path is used. Nigeria and Tanzania are processed locally from existing CSVs using the same morphology logic.

- **Calculated features:** building count, building density, footprint fraction, total area, mean/std building area, size Gini, nearest-neighbor spacing regularity, and orientation entropy.
- **Orthogonal features:** size Gini, spacing regularity, and orientation entropy. Size Gini summarizes inequality in footprint areas inside the H3 resolution-7 cell; spacing regularity is based on the variability of nearest-neighbor distances between building centroids; orientation entropy summarizes whether footprints share a common direction or are more chaotic. Count, density, footprint fraction, and total area are amount features.

OSM roads. OSM road features are computed locally from GeoPackage extracts. Lengths and bearings are computed in a metric CRS, then segments are assigned to H3 resolution-7 cells by midpoint. This midpoint rule is fast and consistent, but it can under-attribute long roads that pass through a cell without their midpoint falling inside it.

- **Calculated features:** road density, road class maximum, segment counts, node counts, intersection counts, dead-end counts, dead-end ratio, degree shares, beta connectivity, grid-like diagnostic, and bearing entropy.
- **Orthogonal features:** dead-end ratio, degree shares, beta connectivity, and bearing entropy. These are topology/form features. Endpoint degree is computed from rounded segment endpoints, beta connectivity uses edges/nodes, and bearing entropy summarizes directional order. The hand-built grid-like diagnostic is dropped from model features so the model sees the raw topology components.

DHS and RWI. DHS outcomes are constructed from HR, IR, KR, and GE files, then aggregated to H3 resolution-7 cells. Outcomes are wealth, improved sanitation, women's education, stunting, and mobile ownership. RWI is parsed and retained for audit completeness but is dropped before training, along with RWI/meta wealth columns, so the models do not use RWI as a shortcut.

Orthogonal feature logic. The orthogonal block is not literally independent of AlphaEarth. It is a deliberately imperfect diagnostic group: demographics, temporal variation, verticality, morphology, and topology are expected to be less reducible to a single visual intensity signal than population count, road length, nightlight mean, or building count. It remains incomplete. It does not yet capture institutions, service access, health systems, food security, disease ecology, or shocks. That incompleteness is useful to keep visible because it helps explain why stunting remains weak and why education/mobile transfer can be calibration-sensitive.

3 Training And Evaluation

3.1 Spatial Splits

Spatial leakage control is handled at the **block level**, not the H3 resolution-7 cell level. Each country is divided into one-degree longitude/latitude buckets, and each H3 resolution-7 cell is assigned to a country-specific spatial block from its grid indices plus a country prefix. This means a test H3 resolution-7 cell is not randomly surrounded by training cells from the same local area.

The fold assignment is greedy and label-aware. Blocks are sorted by labeled H3 resolution-7 load and total H3 resolution-7 load, then assigned to the fold with the lowest combination of label load, country imbalance penalty, and total cell load. Folds are accepted only if each active outcome clears the minimum label gate after buffering. The implemented final gate is 30 labels per fold.

Boundary buffering is applied after fold assignment. Approximate distance to the nearest one-degree block boundary is computed using latitude-adjusted longitude distances, and H3 resolution-7 cells within 5 km of a boundary are marked as buffer cells. The 5 km width is chosen to match the documented DHS displacement distribution: urban clusters are displaced by up to 2 km, and 99% of rural clusters by up to 5 km, with the remaining 1% up to 10 km. A 5 km buffer therefore covers all urban and 99% of rural displacement, while a 10 km buffer (reported for sensitivity in Table 2) is more conservative but discards nearly twice as many labels. Train/test masks then exclude buffered cells by default: training is non-buffer cells outside the held-out fold, and testing is non-buffer cells inside the held-out fold. Coordinates are also excluded from the feature matrix so the models cannot simply learn absolute geography.

For the GNN, the same fold logic matters twice. First, the held-out fold is spatially separated. Second, train and test subgraphs are built so message passing does not directly mix train and test H3 resolution-7 nodes across the split. This is why the graph result can be read as a spatial model rather than a leakage-prone smoothing exercise.

Table 2: Spatial split and buffer accounting. Counts are for the main DHS outcomes unless noted.

Setting	Used	Dropped	Drop %
10 km buffer, 1° blocks	3,466	1,720	33.2%
5 km buffer, stunting	4,219	913	17.8%
5 km buffer, 1° blocks	4,266	920	17.7%

3.2 Models And Why The Comparisons Matter

XGBoost is the main supervised tabular baseline, and it is the model against which the other approaches are evaluated. Gradient-boosted trees provide a strong reference for heterogeneous tabular data of this size: they handle mixed feature scales, nonlinearities, and high-order interactions without requiring a graph. The spatial honesty in this comparison comes from the evaluation design—block-level splits, displacement buffering, and coordinate exclusion—so XGBoost is spatially evaluated even though it does not perform message passing. Comparing the GNN against this baseline therefore helps isolate whether H3 neighborhood aggregation adds predictive value beyond the per-cell features. Fold-safe preprocessing is fitted only on training data: missingness indicators are created from the training split, missing values are filled consistently, and robust scaling is learned on train then applied to test.

The GNN is a single-task GraphSAGE model over H3 resolution-7 adjacency. The original multitask GNN shared one output head across all DHS outcomes and was unstable, especially for mobile and stunting. The final model trains one outcome at a time, adds layer normalization, lowers hidden width, increases dropout, and uses an inner spatial validation split for early stopping.

The fused embedding is trained without DHS labels. It uses separate AlphaEarth and tabular encoders, fuses them into a latent Z of dimension 256, and trains with reconstruction, view-alignment, feature masking, and variance regularization. DHS evaluation happens only afterward through a frozen Ridge probe.

These comparisons are necessary for interpreting the embedding. XGBoost full is the main supervised reference for the current feature table. GNN full asks whether H3 resolution-7 neighborhood aggregation adds signal beyond a strong tabular model. AlphaEarth-only shows the pure EO baseline. The frozen embedding asks whether the combined feature stack can be compressed into a reusable representation without DHS supervision. Its position relative to full XGBoost and AlphaEarth-only indicates how much information survives unsupervised fusion.

All leave-one-country-out results are reported as **pooled** R^2 over concatenated held-out predictions rather than as a mean of per-country R^2. This provides one aggregate measure across all out-of-country predictions. Country-specific estimates remain useful diagnostics because pooled performance can conceal transfer heterogeneity, especially for the smallest countries and weakest outcomes. **LOCO uncertainty intervals are not reported because the evaluation contains only seven independent held-out countries, too few for a stable country-level interval. With seven countries, LOCO should therefore be read as an initial transfer evaluation rather than a precise estimate of performance across a broader country population; expanding the country set is needed for more precise transfer uncertainty.**

For spatial CV, the tables report the mean and standard deviation of R^2 across the five held-out folds. These standard deviations describe fold-to-fold variation; they are not treated as conventional confidence intervals because the five training samples overlap. Where two specifications are compared directly, the discussion also uses paired fold differences computed on the same held-out folds.

4 Results And Discussion

Table 3: XGBoost R^2 by feature set. Spatial-CV entries are mean ± SD across five held-out folds; LOCO entries are pooled across held-out-country predictions.

Outcome	AE only	Tab only	AE+amount	AE+orth.	Full
Panel A: spatial CV, fold mean ± SD					
Wealth	.629±.042	.732±.026	.738±.030	.712±.039	.745±.032
Sanitation	.418±.054	.499±.053	.510±.047	.491±.047	.522±.051
Women edu.	.659±.023	.687±.031	.704±.031	.699±.034	.720±.031
Mobile	.457±.064	.485±.078	.492±.079	.485±.073	.503±.081
Stunting	.201±.027	.184±.023	.204±.024	.214±.019	.214±.023
Panel B: leave-one-country-out, pooled predictions					
Wealth	0.505	0.657	0.682	0.643	0.673
Sanitation	0.281	0.385	0.415	0.387	0.410
Women edu.	0.171	0.329	0.275	0.311	0.306
Mobile	0.004	0.097	0.106	0.085	0.126
Stunting	-0.070	-0.042	-0.030	-0.008	-0.016

Interpretation: A consistent outcome hierarchy emerges. Wealth is the strongest outcome, sanitation and women's education form a middle-to-strong group, mobile ownership is locally predictable but weaker in transfer, and stunting remains near the floor. This ordering is substantively plausible: wealth is strongly expressed in settlement intensity, built form, roads, lighting, and population concentration; sanitation is partly infrastructural and urbanization-linked; women's education is less directly visible but spatially correlated with the same long-run development gradients; mobile ownership is partly infrastructural but also depends on markets, prices, and network roll-out; and stunting is a biological and nutritional outcome that settlement structure should not be expected to capture. The same broad ordering appears across XGBoost, the GNN, and the frozen embedding, suggesting that it is not specific to a single model family.

Wealth provides the strongest initial result.

The pooled LOCO wealth R^2 of 0.673 and pooled spatial-CV R^2 of 0.750 fall within a similar range to established EO-poverty benchmarks. Yeh et al. report pooled held-out-country $R^2 = 0.67$ from public imagery, while Pettersson et al. report approximately 72% held-out-country variance explained using multi-temporal image sequences. This provides an encouraging direction for the augmented AlphaEarth approach using precomputed annual embeddings and tabular geospatial layers. The comparisons remain indicative rather than directly equivalent because of methodological differences, including country coverage, spatial unit, outcome construction, and validation design. Extending the evaluation to a larger country set will show how well this pattern holds more broadly.

The tabular layers add information beyond AlphaEarth alone.

Across the five spatial folds, full XGBoost improves over AlphaEarth-only by a paired mean R^2 difference of $+0.117 \pm 0.013$ for wealth, $+0.104 \pm 0.016$ for sanitation, $+0.061 \pm 0.018$ for women's education, $+0.046 \pm 0.023$ for mobile ownership, and +0.013±0.005 for stunting. Every difference is positive in all five folds, making augmentation over AlphaEarth-only the most consistent comparison in the current evaluation. Pooled LOCO differences are also positive (+0.168, +0.129, +0.135, +0.122, and +0.054, respectively), providing an encouraging initial indication that the added layers may also support transfer. The largest gains occur where population concentration, building amount, roads, and lighting are conceptually close to the target, and the smallest for stunting.

The amount/orthogonal split offers an initial view of where augmentation may come from.

The feature split groups variables by how reducible they are to a single visual-intensity signal: amount features (population, building, light, and road intensity) versus orthogonal features (demographics, temporal variation, verticality, morphology, and topology). AE+amount exceeds AE+orthogonal for wealth by a paired mean R^2 difference of 0.026 ± 0.013 and is higher in all five folds. Sanitation shows a smaller directional difference of 0.018 ± 0.014 and is higher in four of five folds. For women's education and mobile ownership, the mean differences are only 0.004 ± 0.015 and 0.007 ± 0.014, so the two combinations are broadly similar at the present level of fold variability. AE+orthogonal is directionally higher for stunting by 0.010 ± 0.013, also a small difference. These patterns make the grouping useful as an initial diagnostic, while leaving the relative contribution of the two blocks open for a more detailed feature-family ablation.

Table 4: Model comparison. Spatial-CV entries are mean $R^2\pm$ SD across five held-out folds; LOCO values are pooled across held-out-country predictions.

Outcome	XGB full	GNN AE	GNN full	SSL Z
Panel A: spatial CV, fold mean ± SD				
Wealth	.745±.032	.642±.047	.745±.047	.699±.056
Sanitation	.522±.051	.416±.054	.509±.042	.462±.048
Women edu.	.720±.031	.656±.046	.703±.052	.539±.118
Mobile	.503±.081	.375±.117	.524±.086	.315±.137
Stunting	.214±.023	.215±.035	.201±.021	.157±.039
Panel B: leave-one-country-out				
Outcome	XGB R^2	XGB ρ	SSL R^2	SSL ρ
Wealth	0.673	0.832	0.635	0.811
Sanitation	0.410	0.647	0.380	0.637
Women edu.	0.306	0.530	0.184	0.459
Mobile	0.126	0.478	-0.119	0.317
Stunting	-0.016	0.236	-0.095	0.056

GraphSAGE is broadly comparable with XGBoost, without a consistent advantage.

The paired mean GNN-minus-XGB difference is 0.000 ± 0.025 for wealth, -0.012 ± 0.037 for sanitation, -0.017 ± 0.032 for women's education, $+0.021 \pm 0.040$ for mobile ownership, and -0.013 ± 0.025 for stunting. The positive mobile difference is directionally encouraging, but the GNN is higher in only three of five folds; the other outcome comparisons are similarly mixed. The current evidence therefore supports comparable performance rather than a general benefit from message passing. Architecture still matters within the GNN family: the original multitask model was unstable (mobile mean $R^2 = -0.283$, stunting 0.046), while the final single-task, layer-normalized, validation-stopped model raises the corresponding fold means to 0.524 and 0.201. This shows that outcome-specific training substantially improves the graph baseline, even though it does not establish a consistent advantage over XGBoost.

Why a DHS-free augmented embedding, and what the result means.

The regressions above show that fusing AlphaEarth with tabular layers produces useful predictions for several outcomes—but they require DHS labels at training time, and DHS coverage is exactly what is scarce. The motivating question is therefore whether the *representation* itself can be made poverty-aware without consuming labels, so that the expensive, sparse DHS signal is spent only on a light final probe rather than on learning the features. The frozen fused embedding tests this directly: AlphaEarth and tabular encoders are fused into a 256-dimensional latent Z trained only with reconstruction, view-alignment, masking, and variance objectives—never seeing a single DHS value—after which a frozen Ridge probe reads outcomes off Z.

The frozen embedding gives an encouraging initial result for wealth and sanitation. Relative to full XGBoost, its paired spatial-CV difference is -0.047 ± 0.027 for wealth and -0.060 ± 0.012 for sanitation, and it is lower in all five folds. However, it exceeds the supervised AlphaEarth-only XGBoost baseline by 0.070 ± 0.022 for wealth and 0.044 ± 0.015 for sanitation, again in all five folds. A useful portion of the augmentation signal therefore appears to survive compression into a representation trained without DHS labels, although the full supervised feature model remains stronger. Pooled LOCO shows a similar ordering: wealth is 0.635 for the frozen embedding versus 0.673 for full XGBoost, and sanitation is 0.380 versus 0.410. Published image-only estimates provide useful context, but remain indicative rather than directly equivalent because their datasets and evaluation designs differ.

A useful augmented embedding could reduce downstream adaptation for a new survey, indicator, or country to fitting a light probe on a reusable representation, rather than repeating the full feature-learning workflow for every task. The current seven-country results do not establish production readiness, but they provide initial evidence that the approach retains meaningful wealth and sanitation information and warrants evaluation over a larger country and outcome set.

Augmentation is the larger observed increment in the current evaluation.

Pettersson and Daoud report cluster-level wealth R^2 of 0.546 for the AlphaEarth image-only baseline and 0.569 for their best graph variant, with graph structure adding approximately 0.02. In the present results, the paired full-versus-AE-only XGBoost difference for wealth is 0.117 ± 0.013, while the mean GNN-versus-XGB full difference is 0.000 ± 0.025. The two sets of results are broadly compatible with the possibility that augmentation contributes more than graph topology in this setting. This remains an indicative comparison because spatial unit, country set, target construction, and cross-validation design differ across studies.

Wealth: the strongest initial result.

AlphaEarth performs well, while the augmented XGBoost and GNN models both reach a fold-mean R^2 of approximately 0.745. Pooled LOCO remains strong at 0.673 for full XGBoost. Tabular-only is close to full in pooled LOCO (0.657 vs. 0.673), suggesting that the engineered socioeconomic variables contain much of the observed transfer signal, while the consistent full-versus-AE-only gain shows that augmentation still adds information.

Sanitation: medium-strong and one of the better transfer outcomes.

Full XGBoost reaches pooled $R^2 = 0.526$ spatially and 0.410 in LOCO, while the frozen embedding reaches 0.467 and 0.380. The comparatively modest gap is consistent with sanitation being captured partly by broad settlement and infrastructure gradients, although it does not establish which features drive the result. Population-accessibility (gravity) features were already present and added negligibly over population density; travel-time-to-city and travel-time-to-health-service features were not included and are the cleaner accessibility ablation to test next.

Women's education: spatially organized despite not being directly visible.

Spatial CV is high (pooled full XGB 0.723; AE-only 0.663), suggesting that women's education is organized through correlated socioeconomic gradients that are partly represented in

EO and geospatial features. Transfer is weaker but positive in pooled LOCO (0.306), and rank information remains useful even where absolute calibration fails in particular countries. Future work should therefore report ranking and calibration separately.

Mobile ownership: locally predictable, fragile in transfer.

Spatial CV is non-trivial (pooled XGB full 0.516, GNN full 0.541), suggesting that socioeconomic and infrastructure gradients carry signal even though mobile ownership is not a direct built-environment variable. The apparent GNN advantage is small relative to fold variation and is not consistent across all folds. Pooled LOCO is much weaker (0.126), which may reflect country-specific telecom markets, pricing, handset diffusion, and survey timing. The transfer result remains an initial indication rather than evidence of broad generalization.

Stunting: consistently weak in the current evaluation.

Stunting remains around $R^2 = 0.20$ across the supervised spatial-CV models and is near zero or negative in pooled LOCO. AE+orthogonal is directionally higher than AE+amount, but the paired difference is small (0.010 ± 0.013) and should not be treated as a settled ordering. The feature table likely lacks variables that matter more directly for child nutrition—disease ecology, food security, maternal/child health-service access (travel-time-to-health, untested here), shocks, and survey-season controls—so stunting helps mark the current boundary of settlement-structure prediction.

4.1 Toward Explainability

A central question for policy use is whether these models rely on meaningful built-environment signal or on country-level artifacts that happen to correlate with the outcome. Gain-based feature importances from the XGBoost full models give a first, coarse view of which inputs the supervised models actually use, but gain importance reflects only how often and how usefully a feature is split on—not its directional or marginal effect—and it is diluted across the 64 correlated AlphaEarth dimensions.

The intended next step, aligned with the broader project's explainability objective, is to apply model-agnostic attribution (e.g. SHAP) and spatial residual analysis to test whether predictions rest on interpretable built-environment and demographic structure rather than on country identity. This matters most for the transfer (LOCO) setting, where reliance on country-specific artifacts would inflate within-sample performance while failing to generalize.

5 Current Limitations And Next Steps

Country coverage is an initial seven-country evaluation. This supports the present pipeline and transfer analysis, while a larger country set is needed to assess how broadly the findings extend across Africa. The next version should expand coverage once DHS/GPS files are consistent.

Temporal alignment is imperfect. AlphaEarth is available annually through 2024, VIIRS/NDVI can be year-matched more flexibly, but GHSL height is static at 2018 and some sources are top-coded.

The orthogonal/amount split is a first cut. Amount features show a modest, consistent advantage for wealth, while the remaining outcome differences are smaller or more variable. The grouping is coarse and does not yet separate institutions, service access, health systems, food security, disease ecology, or shocks, which may matter most for the weaker outcomes. Refining the grouping and adding genuinely non-visual context is a natural next step.

Service-access features are only partially tested. Population-accessibility (gravity) features were included and were inert. Travel-time-to-city and travel-time-to-health-service features were not included in the reported runs and are the cleanest next accessibility ablation, particularly for sanitation and stunting.

LOCO calibration varies across countries. Pooled results are encouraging for wealth and sanitation, while country-level diagnostics show that absolute calibration can still fail in particular settings. Ranking and calibration should therefore be reported separately.

Uncertainty estimates remain preliminary. Fold-level standard deviations show how spatial-CV performance changes across the five held-out folds, but they are descriptive rather than conventional confidence intervals because the training sets overlap. A paired spatial-block bootstrap would provide a stronger uncertainty estimate for pooled metrics and model differences. For LOCO, expanding beyond seven countries is the most important step toward more precise inference about transfer.

Next steps. Expand countries; run 0/5/10 km buffer sensitivity; add travel-time-to-city and travel-time-to-health-service features as their own ablation (distinct from the population-accessibility features already tested); split feature-family ablations beyond amount/orthogonal; improve the fused embedding with masked feature-group prediction, contrastive AE-tabular alignment, country-robust losses, and graph-aware objectives; adopt a displacement-aware (fuzzy) label scheme in the spirit of Pettersson and Daoud rather than a hard boundary buffer; compare against direct Sentinel-2, Landsat, and higher-resolution imagery where available; add explainability to test whether models rely on meaningful built-environment signals or country artifacts.

Provenance

All reported figures are derived from the same seven-country H3 resolution-7 feature table. Spatial-CV table estimates and paired differences are calculated from `all_metrics_with GNNsingleheadtraining_smoothing.csv`; pooled spatial-CV and LOCO values are computed over concatenated out-of-fold or held-out-country predictions. The final model comparison uses the single-task GNN results.

Linked Papers

1. *Combining satellite imagery and machine learning to predict poverty.* Neal Jean, Marshall Burke, Michael Xie, et al.
2. *Using publicly available satellite imagery and deep learning to understand economic well-being in Africa.* Christopher Yeh, Anthony Perez, Anne Driscoll, et al.
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